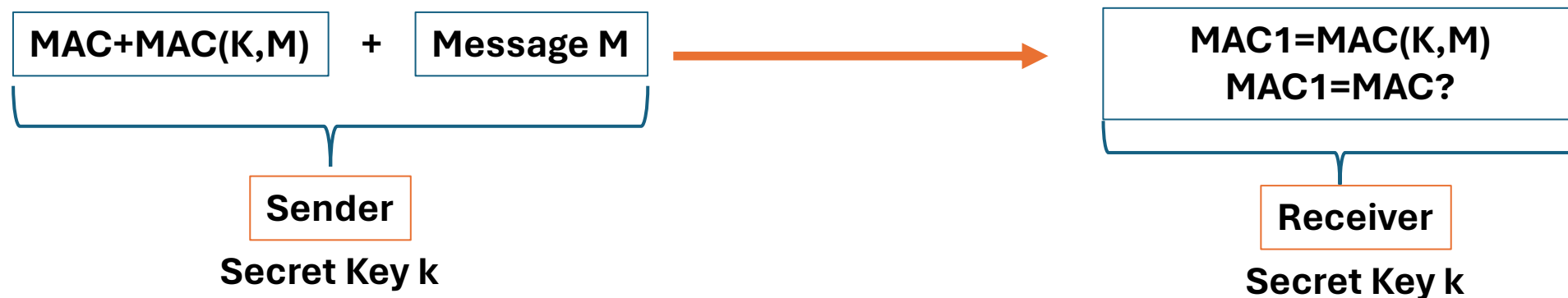


HMAC

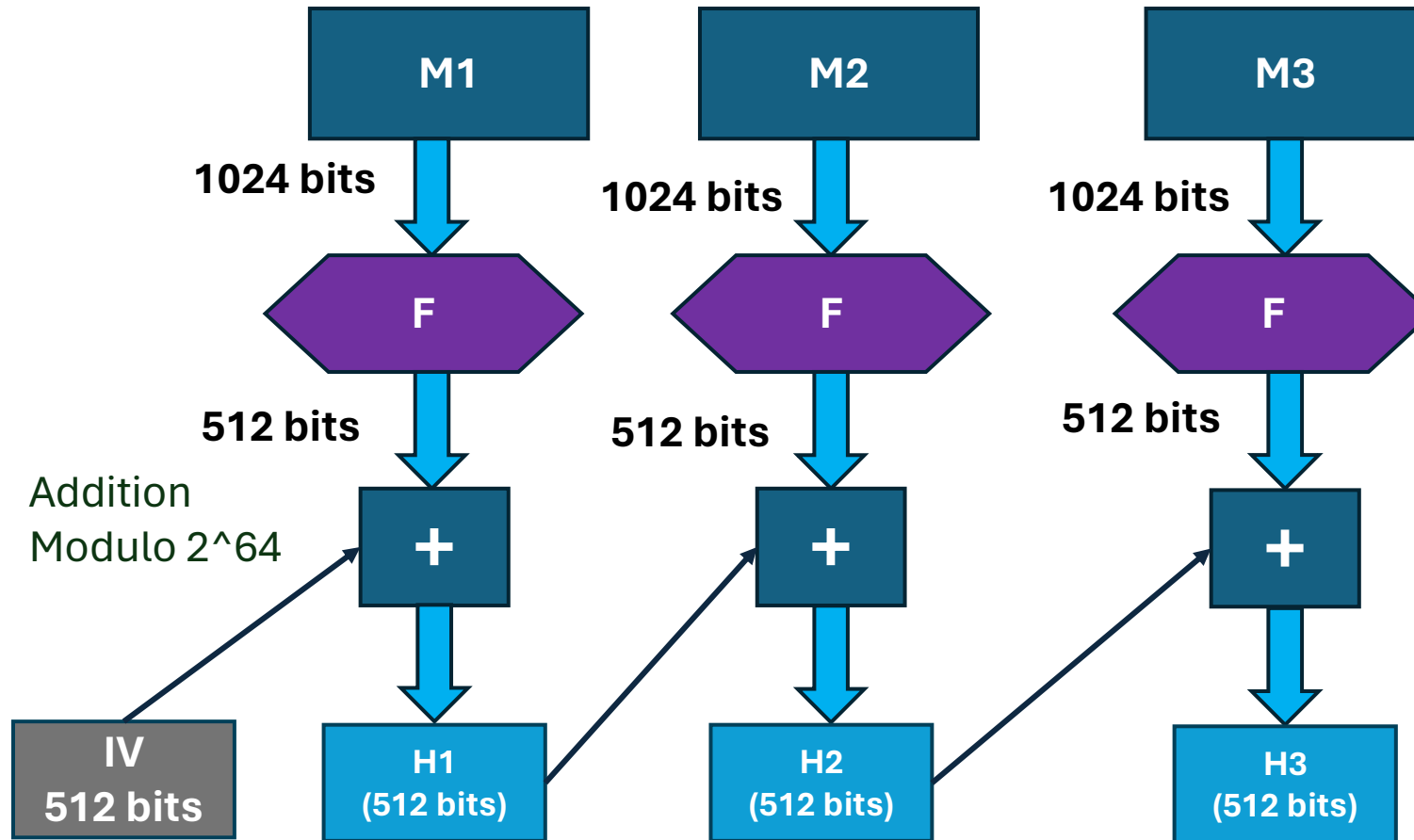
Module IV

MAC & HMAC

- Message authentication ensures Integrity & source legality.
- A MAC is an algorithm that requires the use of a secret key.
- A MAC takes a variable length message & a secret key as Input and produces an authentication code.
- A recipient in possession of the secret key can generate an authentication code to verify the Integrity of the message.

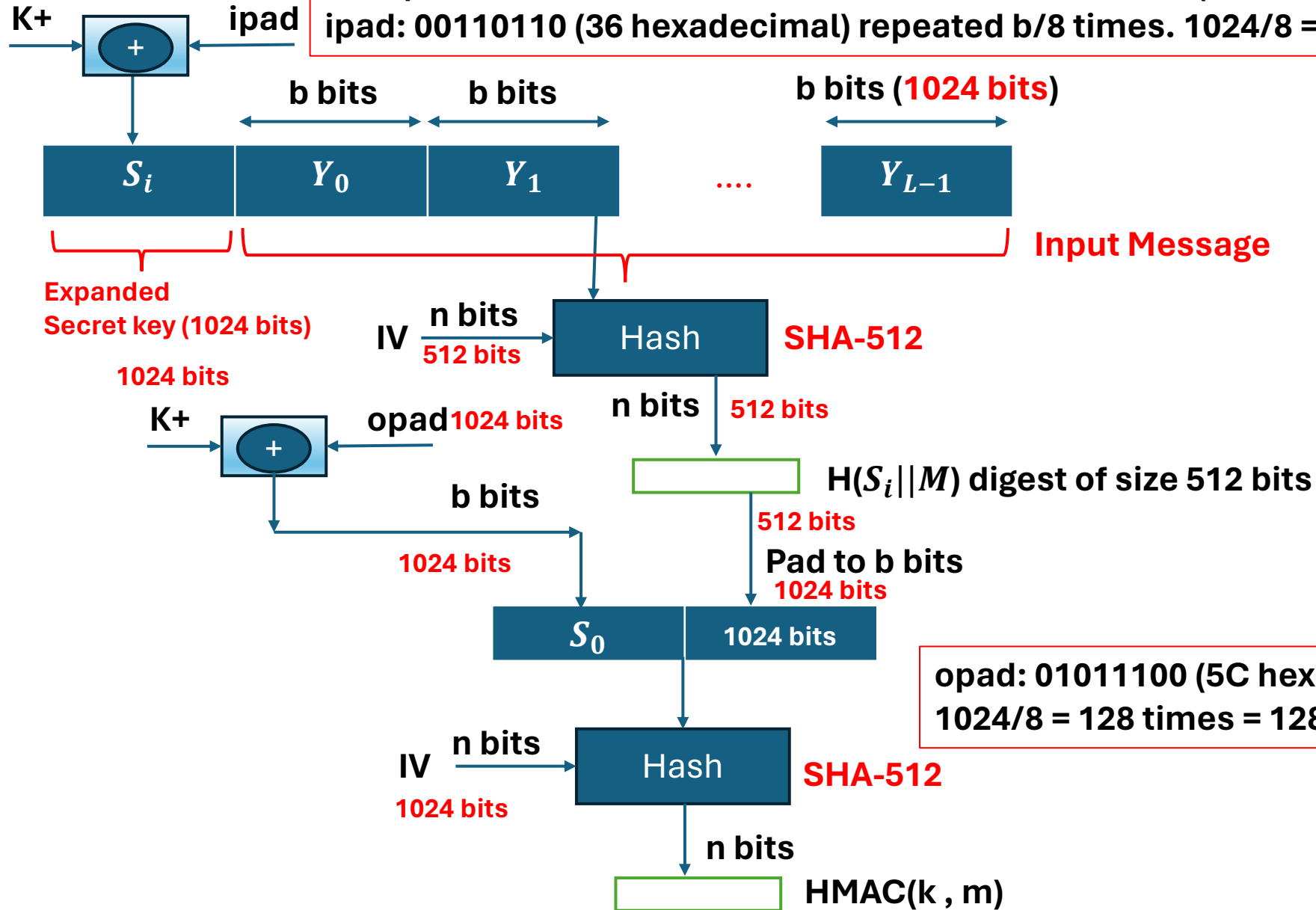


SHA-512 Overall block diagram-recap



HMAC: Message authentication based on hash functions

K+: K padded with zeros on the left so that the result is b bits (1024 in case of SHA-512 in length).
ipad: 00110110 (36 hexadecimal) repeated b/8 times. $1024/8 = 128$ times = $128 \times 8 = 1024$ bits

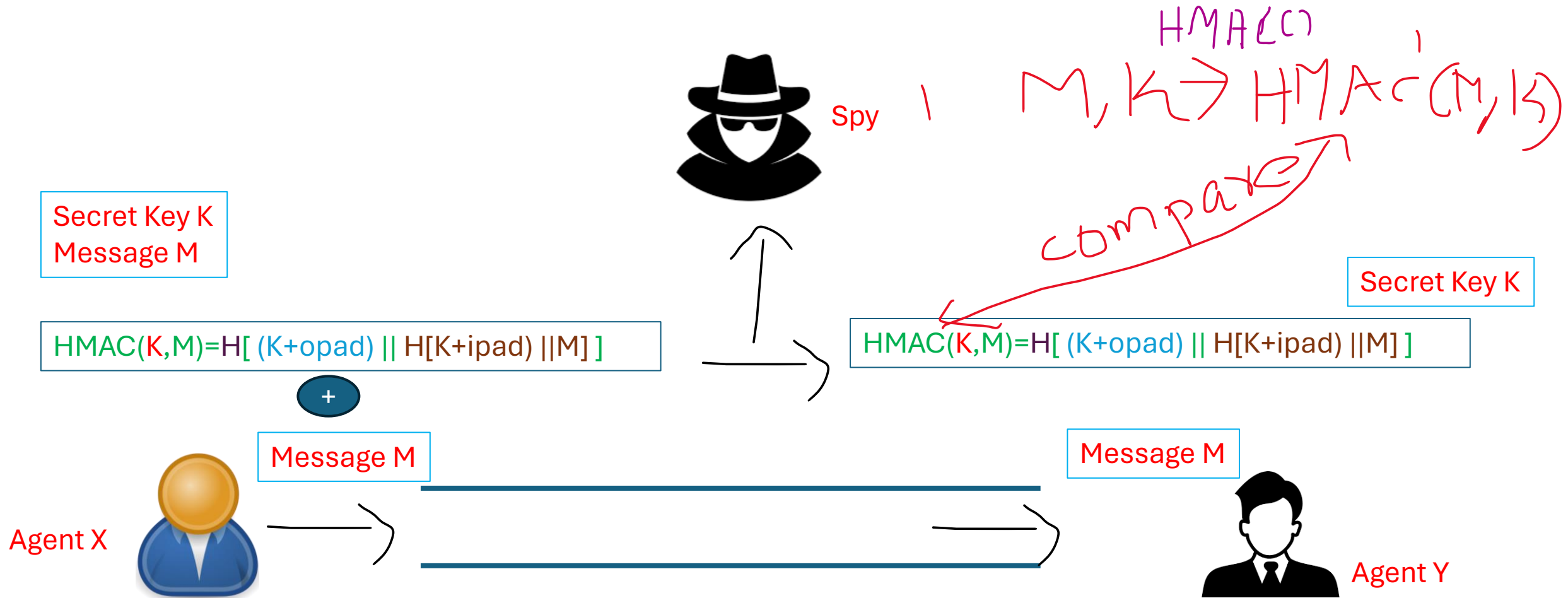


HMAC Structure

- H = embedded hash function (e.g: MD5, SHA-1, RIPEMD-160)
- IV: Initial value input to hash function
- M = Message Input to HMAC (Including the padding specified in the embedded hash function).
- $Y_i = i^{th} \text{ block of } M, 0 \leq i \leq (L - 1)$
- L= Number of blocks in M.
- b=number of bits in a block.
- n=length of hash code produced by embedded hash function.
- K = secret key; recommended length is $\geq n$; if key length is greater than b, the key is Input to the hash function to produce an n-bit key.

Message Authentication Code (HMAC) security

If an Intruder changes Message, it can be Identified by hmac code mismatch.
If an Intruder changes Message as well as hmac code, then it can be Identified by Secret key k.



Algorithm description

1. Append zeros to the left end of K to create a b bit string K^+
2. XOR (bit wise exclusive OR) K^+ with ipad to produce the b -bit block S_i .
3. Append M to S_i .
4. Apply H to the stream generated in step 3.
5. XOR K^+ with opad to produce the b -bit block S_0 .
6. Append the hash result from step 4 to S_0 .
7. Apply H to the stream generated in step 6 and output the result.